EXPERIMENT 17: Word Frequency Distribution Analysis Using CSV

1. AIM

To develop a Python program to load customer feedback from a CSV file, preprocess the text, calculate word frequencies, display the top N most frequent words, and visualize the results using a bar graph.

2. ALGORITHM

1. Import Pandas, Matplotlib and required libraries.
2. Load the data.csv file using Pandas.
3. Read the customer feedback from the feedback column.
4. Convert all text into lowercase.
5. Remove punctuation marks and stop words.
6. Calculate the frequency of each word.
7. Get the value of N from the user.
8. Display the top N most frequent words.
9. Plot a bar graph using Matplotlib.

3. PYTHON CODE

The following is a complete program. It uses a predefined stop-word list, so you don't need to install NLTK.

Experiment 17 – Complete Python Code

```
import pandas as pd

import matplotlib.pyplot as plt

import string

from collections import Counter


# Load CSV dataset

df = pd.read_csv("data.csv")


# Stop words

stop_words = {
```

```
    "the", "and", "is", "are", "a", "an", "to",
    "of", "in", "for", "on", "it", "this", "that",
    "was", "with", "very", "i", "we", "our", "my"
}

# Combine all feedback
text = " ".join(df["feedback"].dropna().astype(str))

# Convert to lowercase
text = text.lower()

# Remove punctuation
text = text.translate(str.maketrans("", "", string.punctuation))

# Split text into words
words = text.split()

# Remove stop words
filtered_words = [
    word for word in words if word not in stop_words
]

# Calculate word frequency
word_frequency = Counter(filtered_words)

# Get N from user
n = int(input("Enter the number of top words: "))
```

```
# Display top N words
top_words = word_frequency.most_common(n)

print("\nTop", n, "Most Frequent Words:")
for word, count in top_words:
    print(word, ":", count)

# Plot bar graph
if top_words:
    words_list = [item[0] for item in top_words]
    counts = [item[1] for item in top_words]

    plt.bar(words_list, counts)
    plt.xlabel("Words")
    plt.ylabel("Frequency")
    plt.title("Top N Most Frequent Words in Customer Feedback")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
else:
    print("No words available to display.")
```

4. INPUT

Create a CSV file named **data.csv** in the same folder as your Python program.

The CSV file should contain:

feedback
The product is very good
Product quality is excellent

feedback

Good product and excellent service

The service is very good

Product quality is good

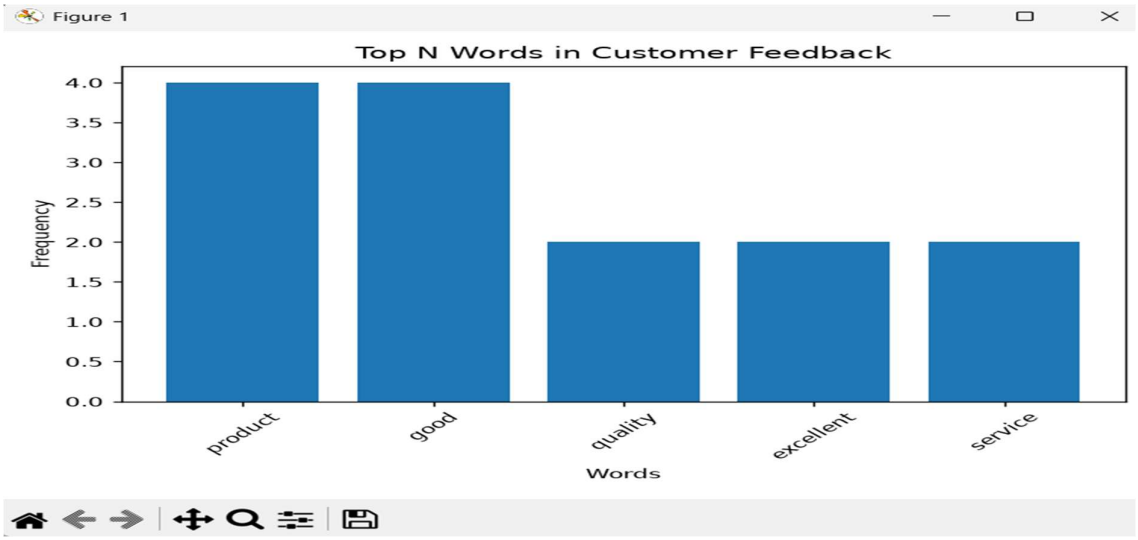
When running the Python program, enter:

Enter the number of top words: 5

5. OUTPUT

```
========== RESTART: C:\Users\arunp\OneDrive\Documents\FDS\EXP 17.py ====
Enter the number of top words: 5

Top 5 Most Frequent Words:
product : 4
good : 4
quality : 2
excellent : 2
service : 2
```



6. RESULT

Thus, the Python program to load customer feedback from a CSV file, preprocess the text, calculate word frequency distribution, display the top N frequent words, and visualize the results using a bar graph was executed successfully.